Beyond Pure Transformers

A Study of Hybrid Mamba-Transformer Architectures in NVIDIA's Nemotron 3 Super

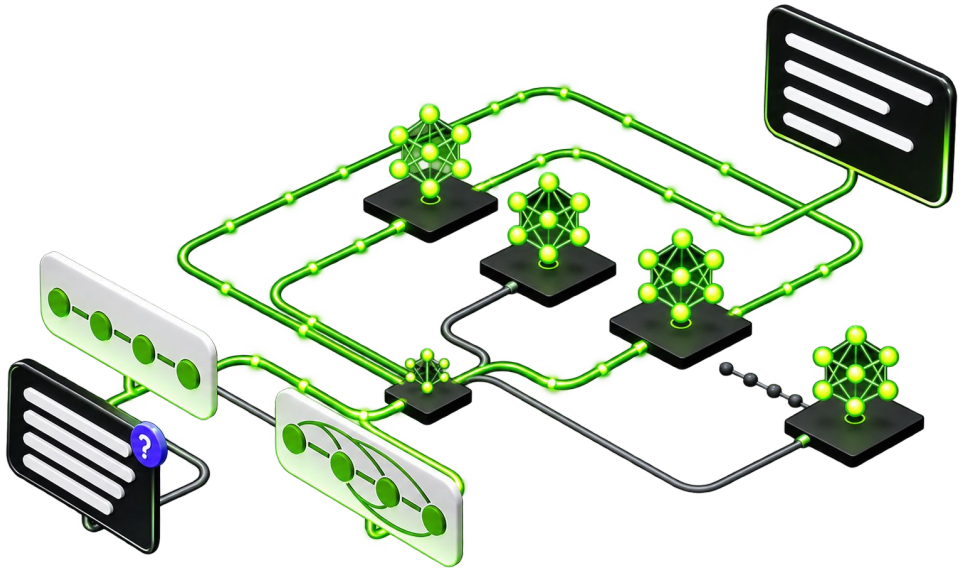
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Objectives

- To understand why Transformers become slow and memory-heavy on large inputs
- To explore Mamba layers, which were proposed to solve that problem
- To study how Nemotron 3 Super combines these layers to develop hybrid model
- To examine how Mixture-of-Experts lets that model stay large while running cheaply
- To highlight what this design suggests about the future of efficient large-scale AI

The Transformer and Its Bottleneck

- Transformer has been the main architecture for most large language models since 2017.
- It uses self-attention to compare all words.
- As the input becomes longer, the amount of computation increases very quickly.
- It stores old words in a Key-Value (KV) cache.
- The KV cache keeps growing as more tokens are added to the input.
- More cache = more GPU memory and slower speed.
- Therefore, pure Transformers become expensive and inefficient for very long-context tasks.

Literature Survey

Study	Main Contribution	Limitation / Research Gap
Dao & Gu (2024) Mamba-2	Introduced Mamba-2, which replaced attention with a small fixed-size memory, making the model 2–8× faster.	Only keeps a summary, so exact details from far back are lost.
NVIDIA (2025) Nemotron 3 Nano	Mixed Mamba with a few attention layers in a 30B model, giving 3.3× higher speed.	Proved only on a small model, using ordinary MoE and normal precision.
NVIDIA (2026) Nemotron 3 Super (paper reviewed)	Scaled the same design to 120B, reading 1M words at 2.2× the speed of GPT-OSS-120B.	Needs large memory, and still behind Qwen3.5 on a few reasoning tests.

Introduction to Nemotron 3 Super

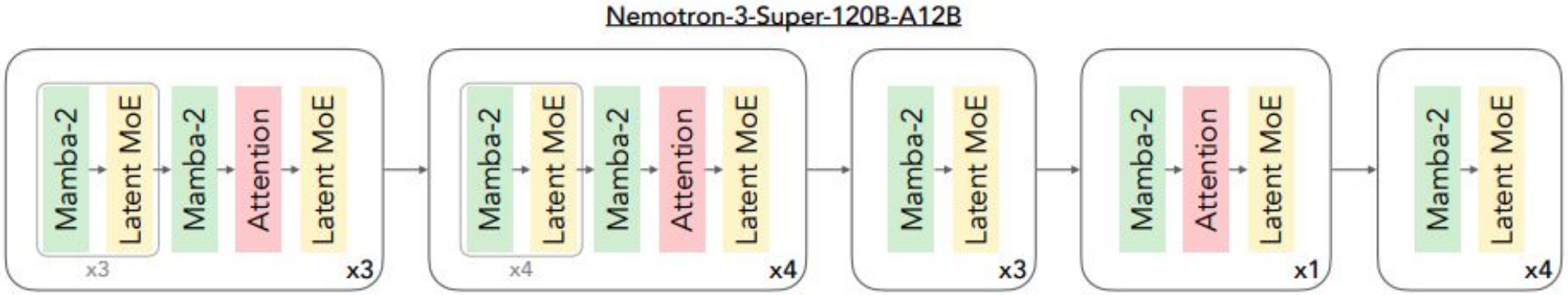
- Nemotron 3 Super is an open AI model developed by NVIDIA in April 2026.
- It has 120 billion parameters, but uses only 12 billion for each token.
- It uses a hybrid architecture with Mamba and Transformer layers.
- It uses Mixture-of-Experts (MoE) to reduce computation.
- It can handle up to 1 million tokens in a single context.
- Runs about 2.2× faster than GPT-OSS-120B at similar accuracy
- Model weights and training data are released openly on HuggingFace

State Space Models and Mamba-2

- A State Space Model reads the text one word at a time, from start to end
- As it reads, it keeps one small note in memory, called the state
- After each word, it updates that note, writing in what matters and dropping what does not
- So the note stays the same size, whether the text is 1,000 or 1,000,000 words
- Mamba-2 is a faster State Space Model used in Nemotron 3 Super.
- Its only weakness is that the note is a summary, so far-back details can fade

Hybrid Interleaved Architecture

- Nemotron 3 Super has 88 layers in total.
- Most layers are Mamba-2 for fast processing.
- A few Transformer attention layers are added between Mamba layers.
- Attention layers help the model connect and understand long-range information.
- So Mamba does the heavy work and attention is used only where it is needed
- This hybrid design makes the model fast and supports 1 million token context.



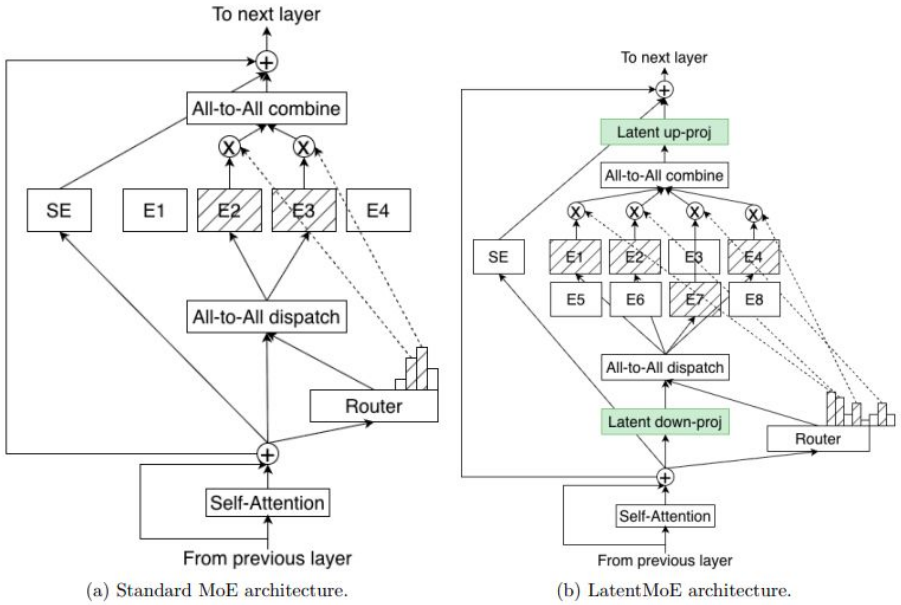
Nemotron 3 Super layer pattern (NVIDIA, 2026, Fig. 2)

Mixture-of-Experts: The Basics

- MoE divides the model into many small experts.
- A small network called the router decides which experts each token should use.
- Only a few experts work at a time — here, 22 out of 512
- The rest stay idle, so the model is huge but the work stays small
- This is why 120B parameters exist, yet only about 12B run for each token
- This saves memory and computation.
- The model stays large, fast, and efficient.

LatentMoE

- Experts normally work with 4096 dimensions, which uses more memory.
- LatentMoE first reduces the token size from 4096 to 1024.
- Experts process the token in this smaller space, reducing memory traffic by 4×.
- After processing, the token is converted back to 4096 dimensions.
- This gives better accuracy with almost the same computation cost.



Standard MoE compared with LatentMoE (NVIDIA, 2026, Fig. 3)

Multi-Token Prediction and Speculative Decoding

- Normally, the model predicts one token at a time.
- MTP predicts multiple future tokens in one step.
- The big model then checks all those guesses in a single run.
- Correct guesses are kept, and wrong ones are generated again.
- Nemotron 3 Super accepts an average of 3.45 tokens in one check.

Category	DSR1	Qwen3 Next	Nemotron3 Super
Coding	2.99	**4.32**	3.78
Humanities	2.67	3.07	**3.26**
Math	2.98	**3.89**	3.73
Multilingual	2.83	3.97	**4.05**
QA	2.63	3.09	**3.16**
RAG	2.79	3.53	**3.78**
Reasoning	2.80	3.47	**3.59**
Roleplay	2.19	2.17	**2.82**
STEM	2.79	**3.37**	3.30
Summarization	2.59	3.06	**3.48**
Writing	2.41	2.69	**2.99**
Average	2.70	3.33	**3.45**

Average acceptance lengths on SPEED-Bench (NVIDIA, 2026, Table 2)

NVFP4 Low-Precision Pretraining

- Normal training uses BF16 (16-bit precision).
- Nemotron 3 Super uses NVFP4 (4-bit precision) for most of its training.
- 4-bit precision reduces memory usage and increases training speed.
- Important layers like Attention, MTP, and Embeddings still use BF16 for accuracy.
- The model was trained on 25 trillion tokens, showing 4-bit training is stable and efficient.

Training Pipeline Overview

- Step 1: Train the model on 25 trillion tokens to learn language.
- Step 2: Train it to handle 1 million token long context.
- Step 3: Fine-tune it with 7 million conversations for better responses.
- Step 4: Train it for coding, math, and tool-use tasks.
- Step 5: Retrain MTP heads to make text generation faster.



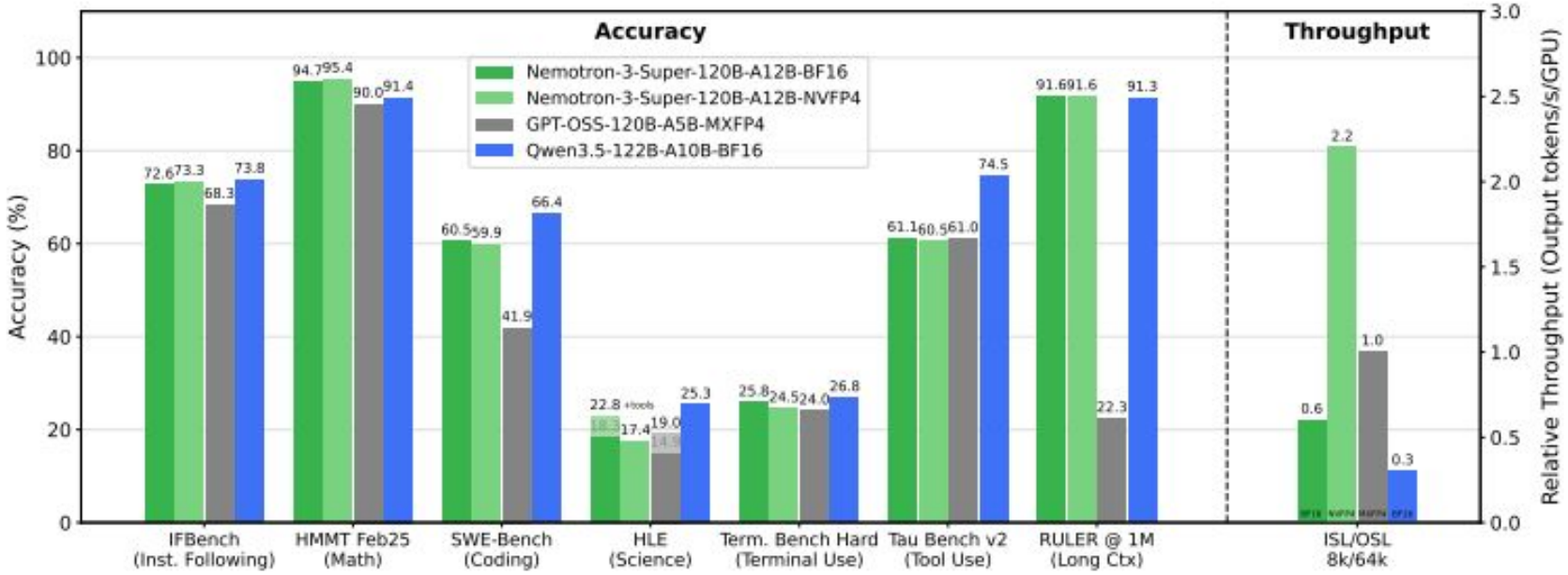
Post-training pipeline (NVIDIA, 2026, Fig. 12)

Quantization for Deployment

- After training, the model is compressed to run faster and use less memory.
- Two compressed versions are released, one for older GPUs and one for the newest ones
- AutoQuantize selects the best precision for each layer.
- Compression caused rounding errors in the Mamba state, making answers longer.
- A random rounding method fixed the problem and kept 99.8% accuracy.

Results and Benchmarks

- Compared against GPT-OSS-120B and Qwen3.5-122B on accuracy and throughput
- Matches or beats GPT-OSS-120B on accuracy, trailing Qwen3.5 slightly on some reasoning tasks
- Runs 2.2× faster than GPT-OSS-120B and 7.5× faster than Qwen3.5-122B
- On RULER at 1 million tokens it scores 91.6, while GPT-OSS-120B scores only 22.3



Accuracy and throughput comparison (NVIDIA, 2026, Fig. 1)

Applications

- Coding agents can read and fix GitHub projects.
- AI assistants can run terminal commands and multi-step tasks.
- Tool-using agents can use APIs, search the web, and answer customer queries.
- Long-document analysis for contracts, research papers, and financial reports.
- Companies can fine-tune and run the model on their own hardware.

Advantages and Limitations

Advantages

- Supports 1 million token long context.
- Runs 2.2× to 7.5× faster than comparable models at similar accuracy
- Only 12B of 120B parameters work per token, keeping compute low
- Weights and datasets are released openly for anyone to use

Limitations

- All 120B parameters must still be held in memory, so hardware needs stay high
- Expert routing makes serving and deployment more complex than a dense model
- Compressing the Mamba state causes rounding errors to build up over long outputs
- Still behind Qwen3.5-122B on some reasoning benchmarks.

Conclusion

- Pure Transformers become slow and memory-heavy for long inputs.
- Nemotron 3 Super combines Mamba and Attention in a hybrid architecture.
- It uses 120B parameters, but only 12B are active for each token.
- LatentMoE and Multi-Token Prediction improve speed and efficiency.
- Overall, Nemotron 3 Super is a fast, efficient, and long-context AI model.

References

[1] NVIDIA, “Nemotron 3 Super: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning,” arXiv:2604.12374, 2026.

[2] NVIDIA, “Nemotron 3 Nano: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning,” arXiv:2512.20848, 2025.

[3] T. Dao and A. Gu, “Transformers are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality,” arXiv:2405.21060, 2024.

[4] V. Elango et al., “LatentMoE: Toward Optimal Accuracy per FLOP and Parameter in Mixture of Experts,” arXiv:2601.18089, 2026.

[5] F. Gloeckle et al., “Better & Faster Large Language Models via Multi-Token Prediction,” ICML, 2024.

[6] D. Lepikhin et al., “GShard: Scaling Giant Models with Conditional Computation and Automatic Sharding,” arXiv:2006.16668, 2020.